

Quantum Science & Technology Across Africa



INTERNATIONAL YEAR OF
Quantum Science
and Technology

Workshop to forge an African network in Quantum Science & Technology (QST)

- Get overview of QST activities in Africa
- Build strong African QST network to connect with networks in Germany and globally
- Initiate collaborations attracting Industry investments

23-27

JUNE 2025

Durban
South Africa

CQCtec

Centre for Quantum
Computing & Technology

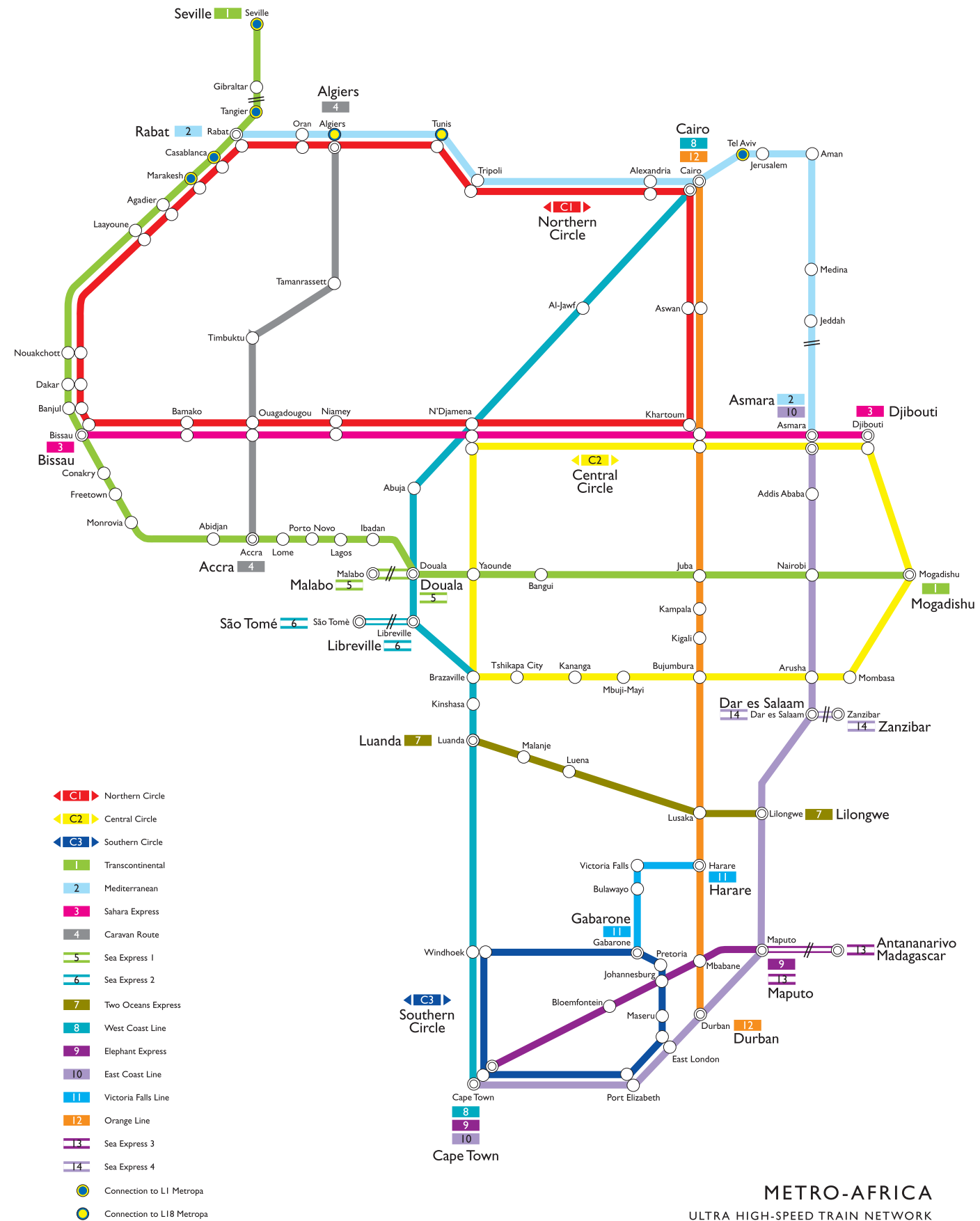
WILHELM UND ELSE
HERAUS-STIFTUNG



South African
Quantum
Technology
Initiative



CONNECT | CO-ORDINATE | CONTRIBUTE



METRO-AFRICA

ULTRA HIGH-SPEED TRAIN NETWORK

Thank you to the sponsors!

Thank you to the sponsors!

**WILHELM UND ELSE
HERAEUS-STIFTUNG**



Thank you to the sponsors!

**WILHELM UND ELSE
HERAEUS-STIFTUNG**



**UNIVERSITY OF
KWAZULU-NATAL** TM

**INYUVESI
YAKWAZULU-NATALI**



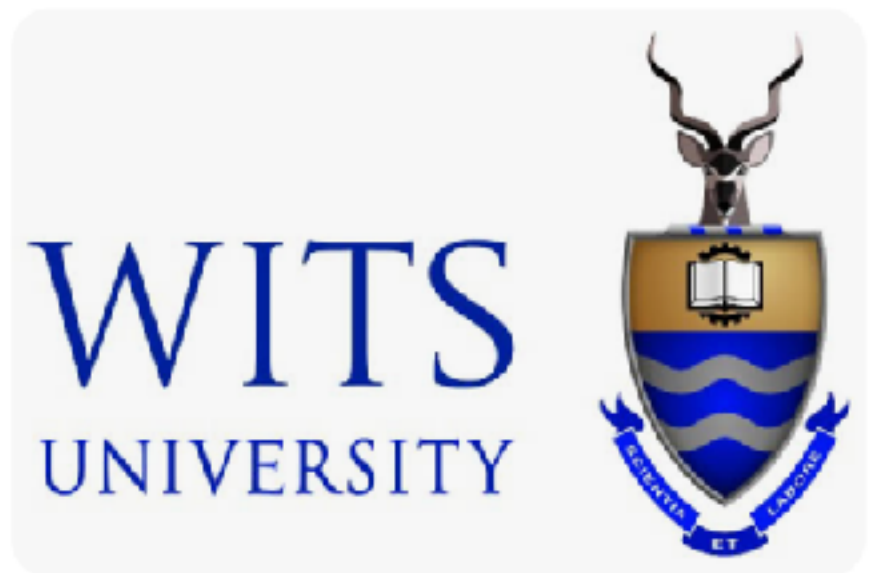
Thank you to the sponsors!

**WILHELM UND ELSE
HERAEUS-STIFTUNG**



**UNIVERSITY OF
KWAZULU-NATAL** TM

**INYUVESI
YAKWAZULU-NATALI**





**South African
Quantum
Technology
Initiative**

Thank you to the sponsors!



**South African
Quantum
Technology
Initiative**

Workshop commitments

- During the workshop accommodation and meals with cold drinks are payed for, thanks to the sponsors. The rest depends on you.
- Talks should not exceed 25 min to leave 5 min for questions
- Reliable Communication: If in doubt, please ask!

Motivation

- Quantum Technology threatens to deepen the gap between *rich* and *poor* countries.
- Quantum Technology presents a chance for development.
- Development needs Industry!
- Industry needs labs and experiments to create expertise to design and build devices.
- Can we use QST for development across Africa?



Quantum Technology Monitor 2024 overview (1/2)

Private and corporate funding for QT start-ups slowed from prior years, while a strong flow of public funding was announced by several governments, totaling ~\$42 billion. The ecosystem continues to progress toward unlocking an estimated economic value of ~\$2 trillion by 2035.

Investments and ecosystem

\$8.5B

+25% YOY

total cumulative global QT start-up investment

367

start-ups in the QT ecosystem

+5% YOY

\$42B

+26% YOY

total government investment announced

+XX% Compared to previous year

Quantum technology market size scenarios for 2035 and 2040

Based on existing development road maps and assumed adoption curve

	Quantum computing	Quantum communication	Quantum sensing
2035	\$28B–\$72B	\$11B–\$15B	\$0.5B–\$2.7B
2040	\$45B–\$131B	\$24B–\$36B	\$1B–\$6B



Potential economic value from quantum computing in 2035

~\$0.9T–\$2T

potential economic value across four industries by 2035: chemicals, life sciences, finance, and mobility¹



¹Economic value is defined as the additional revenue and saved costs that the application of QC can unlock in all industries. These four industries are the most likely to realize this value earlier than other industries; therefore, they are examined in more depth.

Vision

- Develop QST, in particular identify possibilities to develop experimental capabilities in Africa
- Form an accountable QST network across the African continent to attract funds from e.g. the AU
- Get advise from experts from other networks and collaborate. Start with the German QST. This is mutually beneficial and furthers the understanding and cohesion between people!



**First African
Quantum
Technology
Symposium**

CQCtec

Centre for Quantum
Computing & Technology



**INTERNATIONAL YEAR OF
Quantum Science
and Technology**



**QST
CO-ORDINATION
CONFERENCE**



G20

Aims

1. **to bring together leading researchers and experienced research administrators from Africa and from DPG, WEH and ICTP;**
2. **to map out the existing QST higher education and research landscape** across the continent;
3. **to map out existing interfaces between QST research and industrial use-cases, to the benefit of African economies;**
4. **to identify elementary needs** of the diverse players in QST across the continent, **as well as impediments for sustainable progress;**
5. **to identify potential, realistic roles of non-African partners,** through a dedicated effort to define realistic short- and mid-term goals.
6. **To establish a sustained dialogue on QST across the continent,** with the involvement of specifically targeted **partners from public administration, industry and learned societies,** and involving established as well as **young researchers and students at the graduate level;**
7. **to set-up a Steering Committee** and a slim and easily sustainable web-platform for the continuation of the discourse seeded by the Durban meeting.

“If you try to solve a problem and you arrive at a solution that looks difficult, then you have not understood the problem.”

–my interpretation of Cornelius Fai’s words yesterday evening

“If you try to solve a problem and you arrive at a solution that looks difficult, then you have not understood the problem.”

–my interpretation of Cornelius Fai’s words yesterday evening

“It always seems impossible until its done.”

“If you try to solve a problem and you arrive at a solution that looks difficult, then you have not understood the problem.”

–my interpretation of Cornelius Fai’s words yesterday evening

“It always seems impossible until its done.”

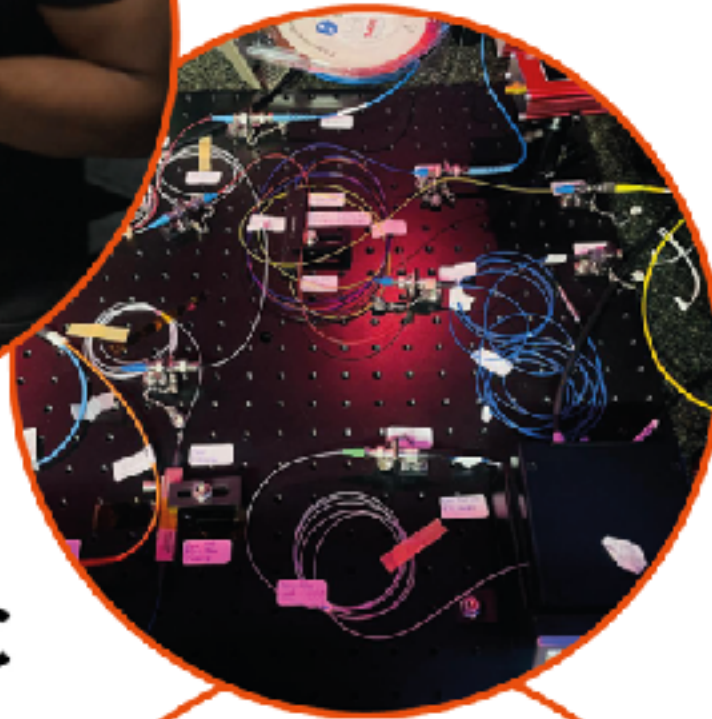
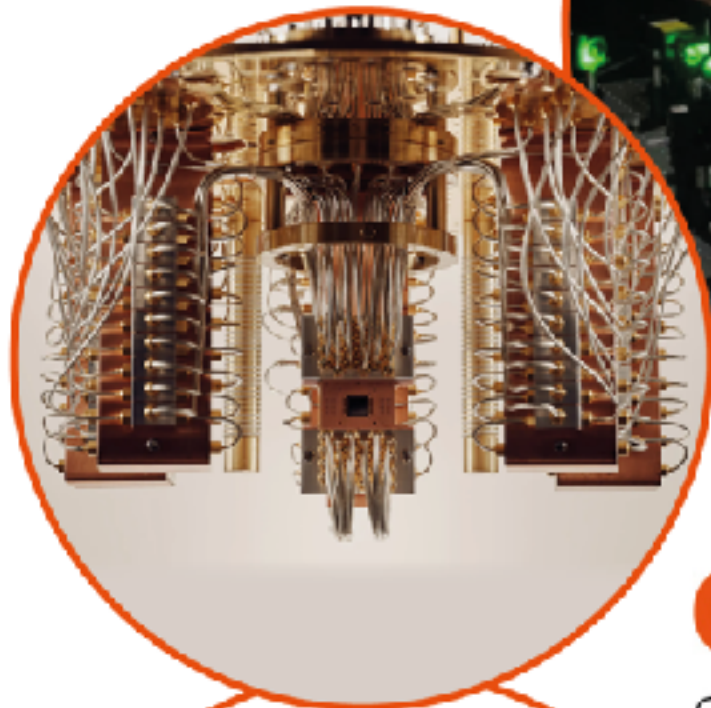
Nelson Mandela



cqctec

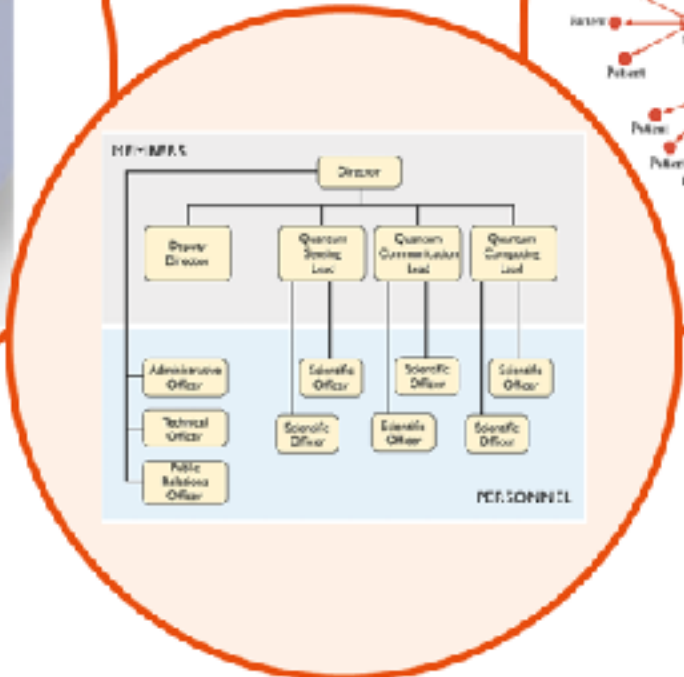
Centre for Quantum
Computing & Technology





cqctec

Centre for Quantum Computing & Technology



Members of the Centre for Quantum Computing and Technology



Prof T. Konrad



Dr. M. Mariola



Prof A. Matthews



Dr. M. Moodley



Dr. A. Pillay



Prof U. Scharler



Prof I. Sinayskly



Prof J. Tapamo



Dr. A. Ukpung



Dr. S. Viriri

Activities

Find use cases for **Quantum Computing**

- optimization (energy systems, logistics, finance),
- drug design
- chemistry and materials,
- pattern recognition and classification.

Activities

Find use cases for **Quantum Computing**

- optimization (energy systems, logistics, finance),
- drug design
- chemistry and materials,
- pattern recognition and classification.



see. think. act.



English

Global



Welcome to ZF

ZF is a global technology company supplying advanced mobility products and systems for passenger cars, commercial vehicles and industrial technology. Its comprehensive product range is primarily aimed at vehicle manufacturers, mobility providers and start-up companies in the fields of transportation and



Activities

Find use cases for **Quantum Computing**

- optimization (energy systems, logistics, finance),
- drug design
- chemistry and materials,
- pattern recognition and classification.

Activities

Find use cases for **Quantum Computing**

- optimization (energy systems, logistics, finance),
- drug design
- chemistry and materials,
- pattern recognition and classification.

Quantum Communication

- set up quantum key distribution (QKD) system in the lab
- design and develop QKD device for commercialization with industry partner.

Activities

Quantum Communication

- set up quantum key distribution (QKD) system in the lab
- design and develop QKD device for commercialization with industry partner.

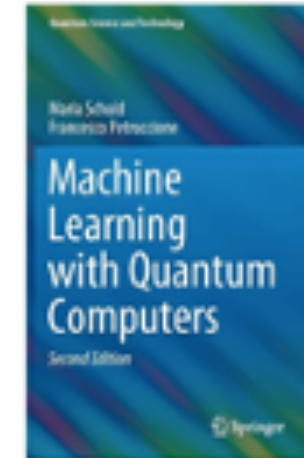
Activities



Quantum Communication

- set up quantum key distribution (QKD) system in the lab
- design and develop QKD device for commercialization with industry partner.

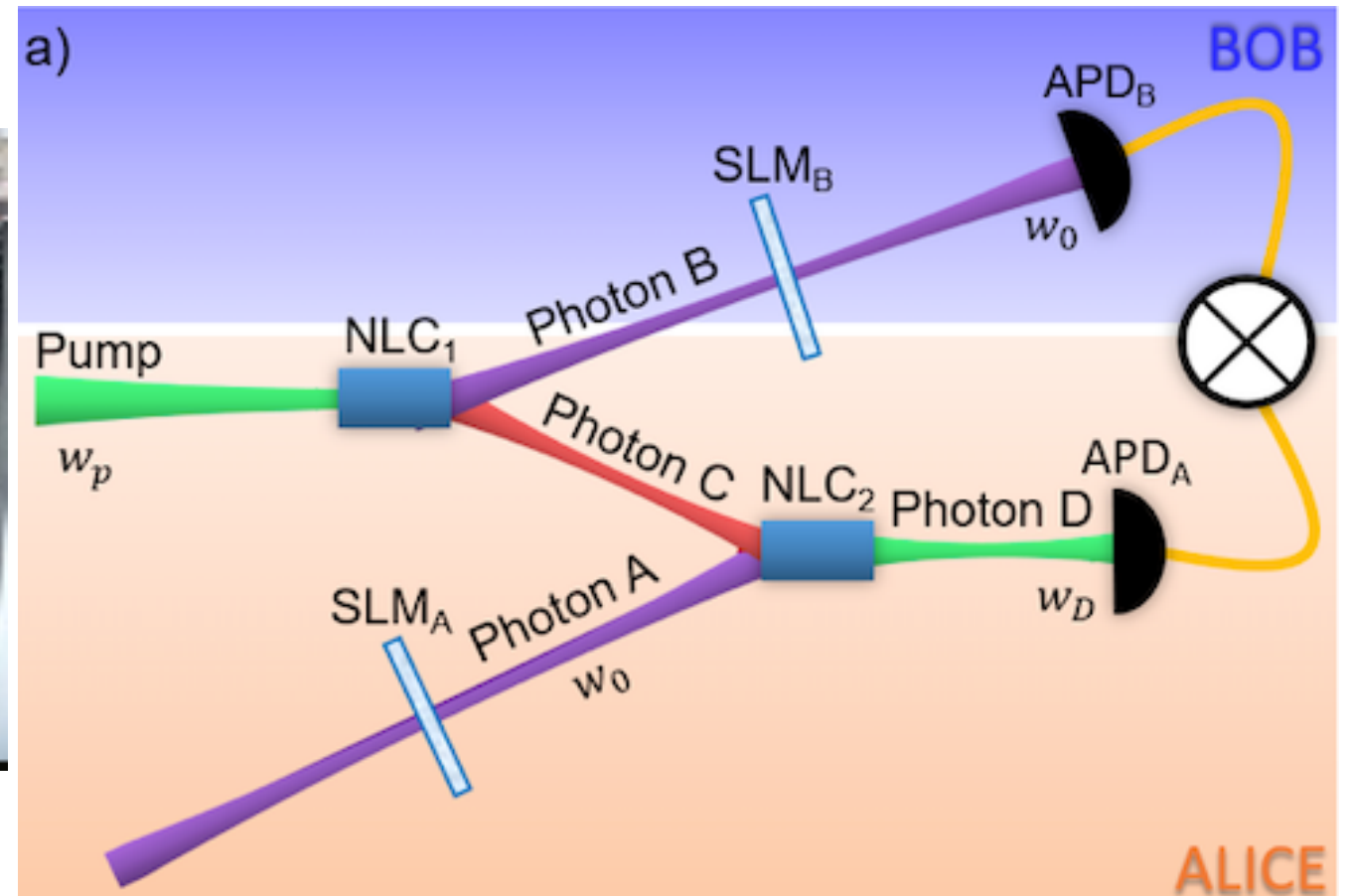
Activities



Quantum Communication

- set up quantum key distribution (QKD) system in the lab
- design and develop QKD device for commercialization with industry partner.

Activities

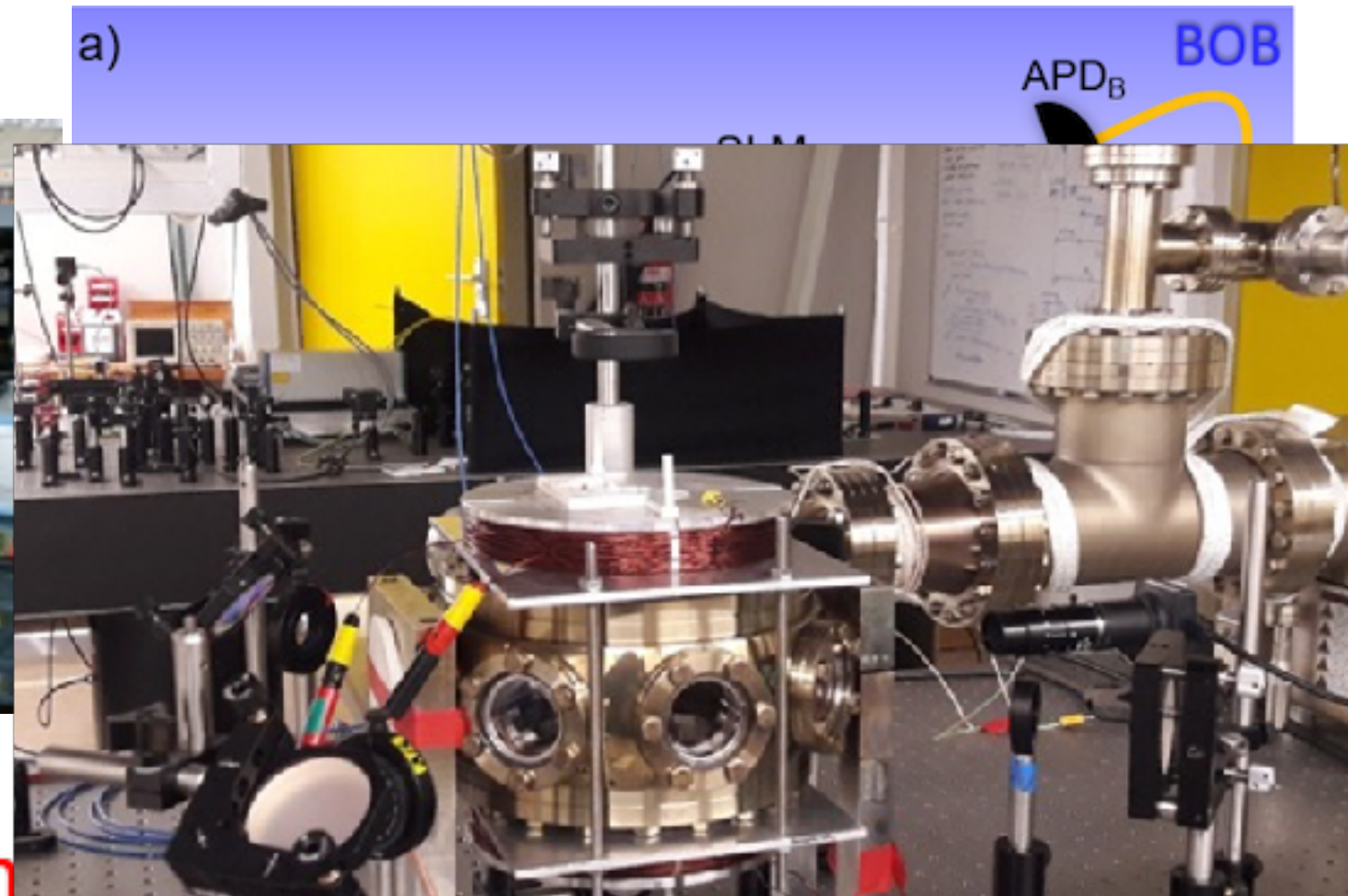


Quantum Communication

#Top25: Nature Communications 14, 8243 (2023)

- set up quantum key distribution (QKD) system in the lab
- design and develop QKD device for commercialization with industry partner.

Activities



Quantum Communication

- set up quantum key distribution (QKD) system in the lab
- design and develop QKD device for commercialization with industry partner.

Cold Atoms lab and Quantum Optics labs

Thank you!

Thank you!



Start now

CQCtec
Centre for Quantum
Computing & Technology

RESEARCH & INDUSTRY ENGAGEMENT: Quantum Technology

NITheCS
National Institute for
Theoretical and Computational Sciences

